



RESEARCH ARTICLE

Role-Based Access Control for Secure Retrieval of Triple-Watermarked DICOM Medical Images

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Abstract

The rapid adoption of digital medical imaging has increased the need for secure storage, transmission, retrieval, and controlled access to medical images. Digital Imaging and Communications in Medicine (DICOM) is widely used for storing and exchanging medical images together with associated patient and clinical information. However, unauthorized access, modification, duplication, and disclosure of sensitive medical information remain important security concerns. This paper proposes a role-based access control framework for the secure retrieval of triple-watermarked DICOM medical images. The proposed framework combines Discrete Wavelet Transform (DWT) and Singular Value Decomposition (SVD) with Role-Based Access Control (RBAC). Three independent watermarks are embedded into selected DWT-SVD coefficients to provide authentication, integrity verification, and ownership protection. RBAC is then applied to control which watermark or associated information can be retrieved by different categories of authorized users. Doctors, radiologists, technicians, researchers, and administrators are assigned different permissions according to their operational requirements. The framework separates authentication from authorization and limits sensitive information exposure according to user roles. Performance can be evaluated using Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), Normalized Correlation (NC), Bit Error Rate (BER), extraction accuracy, and execution time. The proposed approach provides a combined mechanism for image protection and controlled information retrieval while maintaining the visual quality of DICOM images.

Keywords: DICOM; Triple Watermarking; DWT; SVD; Role-Based Access Control; Medical Image Security; Authentication; Integrity; Secure Retrieval

1.0 Introduction

Medical imaging systems have become an essential component of modern healthcare because they allow clinicians to acquire, store, exchange, and analyze patient information in digital form. Modalities such as Magnetic Resonance Imaging (MRI), Computed Tomography (CT), X-ray, and ultrasound generate large amounts of digital medical data. DICOM provides a standardized mechanism for handling medical images and related metadata across different healthcare systems. Although digital imaging improves accessibility and clinical efficiency, it also creates security challenges because medical images contain highly sensitive patient information that must be protected against unauthorized access and manipulation [1]. Traditional protection mechanisms such as passwords, encryption, and network authentication can restrict access to medical data, but they do not always provide a mechanism for verifying whether the image itself has been modified. Medical images may pass through multiple systems during acquisition, diagnosis, storage, and transmission. During these operations, unauthorized modification or replacement of an image may remain difficult to detect. Digital watermarking has therefore been investigated as an additional protection mechanism because information related to authentication, ownership, or integrity can be embedded directly into an image [2].

Transform-domain watermarking methods are generally preferred for medical-image protection because they can provide a suitable balance between imperceptibility and robustness. DWT decomposes an image into

different frequency sub-bands, allowing watermark information to be inserted into selected coefficients. SVD can further improve the stability of the embedding process because small modifications to singular values can produce controlled changes in the image [3]. The combination of DWT and SVD has consequently been explored for robust medical-image watermarking and authentication. However, a major limitation of conventional watermarking systems is that the same embedded information may become available to every authorized user after successful extraction. In a healthcare environment, different users do not necessarily require the same information. For example, a doctor may require patient identification and authentication information, while a researcher may only require anonymized information. A technician may need operational information but should not automatically obtain confidential patient details. Therefore, image protection should be combined with a mechanism that controls information retrieval according to the user's role.

Role-Based Access Control provides an appropriate approach for this requirement. In RBAC, permissions are associated with roles rather than directly with individual users. Users are assigned roles and obtain permissions through those roles [4]. This reduces the complexity of managing permissions in environments containing many users. Integrating RBAC with watermark-based security can therefore provide two complementary protection levels: watermarking protects the integrity and authenticity of the image, while RBAC controls which protected information can be retrieved. The main contribution of this paper is the development of a role-based retrieval mechanism for triple-watermarked DICOM medical images. Three watermarks are considered for different security purposes, such as authentication, integrity verification, and ownership or institutional identification. DWT-SVD is used as the underlying embedding technique, while RBAC determines the extraction privileges associated with each user category. The proposed framework is designed to preserve image quality while reducing unnecessary disclosure of embedded information.

2.0 Literature Review

Medical image security has received considerable attention because medical images are increasingly transmitted through hospital networks, cloud platforms, telemedicine systems, and medical information systems. Existing studies have investigated encryption, digital watermarking, authentication, access control, and hybrid security mechanisms as solutions to protect medical images. Watermarking is particularly useful because it can associate security information directly with the image, allowing authentication or integrity verification even after the image has been transferred between systems [5]. DWT-based watermarking has been widely studied because wavelet decomposition provides multiple frequency components and allows the embedding process to be adjusted according to the characteristics of different sub-bands. Low-frequency components generally contain significant visual information, whereas high-frequency components contain finer image details. Appropriate selection of wavelet coefficients can therefore improve the balance between watermark robustness and visual imperceptibility [6].

SVD-based watermarking is another important transform-domain approach. SVD represents an image matrix using singular values and orthogonal matrices. Singular values are relatively stable under small image modifications, making them useful for watermark embedding. Several hybrid approaches have combined DWT and SVD to obtain improved robustness compared with using a single transform. Such methods have been evaluated against common attacks including compression, noise, filtering, cropping, and other image-processing operations [7]. Multiple-watermark approaches extend conventional watermarking by embedding more than one type of information into a single image. Triple watermarking can be used to separate different security objectives, for example, ownership identification, patient authentication, and integrity verification. This separation allows individual security functions to be evaluated independently. However, multiple-watermark systems also need careful coefficient selection because excessive embedding may reduce image quality [8]. Access-control research has traditionally focused on protecting information resources through authentication and authorization. RBAC is particularly suitable for healthcare systems because medical organizations contain well-defined categories of users with different responsibilities. Role-based policies can restrict users according to their professional requirements and can simplify permission administration [9]. Healthcare access-control research has also emphasized the need for fine-grained authorization because medical information contains different levels of sensitivity.

Recent healthcare-security approaches have increasingly considered combining data protection with access-control mechanisms. However, many watermarking systems concentrate primarily on embedding robustness and image quality, while access-control mechanisms are often implemented independently from the embedded

information. A gap therefore exists in integrating multi-watermark extraction with role-specific authorization. The proposed work addresses this gap by assigning retrieval permissions to the three embedded watermark layers and allowing the system to release only the information permitted by the authenticated user's role.

3.0 Proposed Methodology

The proposed framework consists of five major stages: DICOM image acquisition, triple-watermark preparation, DWT-SVD watermark embedding, user authentication and role assignment, and role-based watermark retrieval.

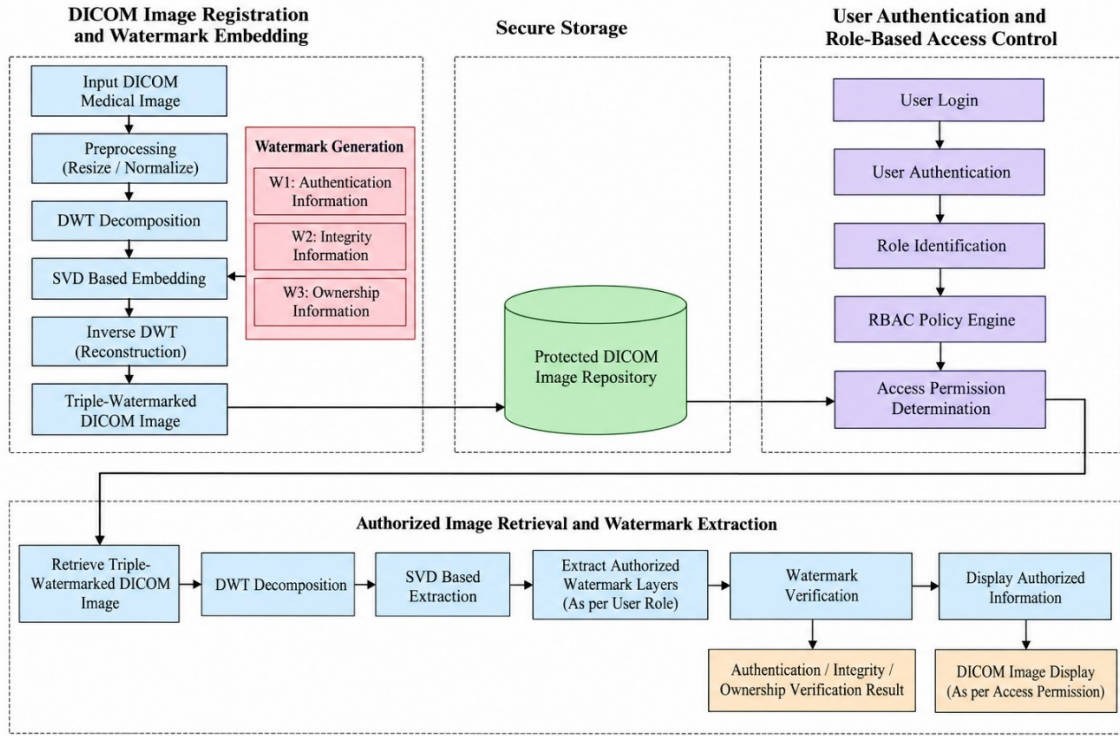


Figure 1. Proposed Role-Based Access Control Framework for Triple-Watermarked DICOM Medical Images

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The DICOM image is first acquired from the medical imaging system. Patient-related metadata that should not be directly exposed can be separated or protected according to the implementation requirements. Three watermark sequences are generated. Watermark 1 can represent authentication information, Watermark 2 can represent integrity-related information, and Watermark 3 can represent ownership or institutional information. The DWT transformation is then applied to the selected image region. The resulting frequency sub-bands are processed using SVD. Watermark information is embedded by modifying selected singular values according to a controlled embedding strength. After inverse transformation, the watermarked image is reconstructed and stored as a protected DICOM image.

During retrieval, the user first provides authentication credentials. After successful authentication, the system determines the user's assigned role. The RBAC policy engine checks the permissions associated with that role. Only the watermark information permitted for that role is extracted.

3.1 Triple-Watermark Structure

The proposed system uses three logically independent watermark layers.

Table 1. Functional Structure of the Three Watermark Layers

Watermark	Main Purpose	Example Information	Security Function
W1	Authentication	Patient/image identification code	Authentication
W2	Integrity	Hash or verification code	Tamper detection

Watermark	Main Purpose	Example Information	Security Function
W3	Ownership	Hospital/department identifier	Ownership verification

The separation of watermark functions provides flexibility because the system does not need to expose all embedded information to every user.

4.0 DWT-SVD Based Watermark Embedding

The proposed watermarking mechanism uses a hybrid DWT-SVD approach. Let the original grayscale medical image be represented by I . The two-dimensional DWT operation decomposes the image into four sub-bands:

$$I \rightarrow LL, LH, HL, HH$$

Where LL represents the low-frequency approximation component, while LH , HL , and HH represent horizontal, vertical, and diagonal detail components, respectively. A suitable sub-band is selected according to the required trade-off between robustness and imperceptibility. SVD is subsequently applied to the selected coefficient matrix:

$$A = USV^T$$

where U and V are orthogonal matrices and SS is a diagonal matrix containing singular values. For watermark W_k , the singular values can be modified using an embedding relation such as:

where represents the modified singular-value matrix and α_k represents the embedding strength. Different embedding strengths can be assigned to the three watermark layers according to their security requirements. The modified matrix is reconstructed using inverse SVD, followed by inverse DWT. The resulting image contains the three embedded watermark layers while maintaining the diagnostic image content.

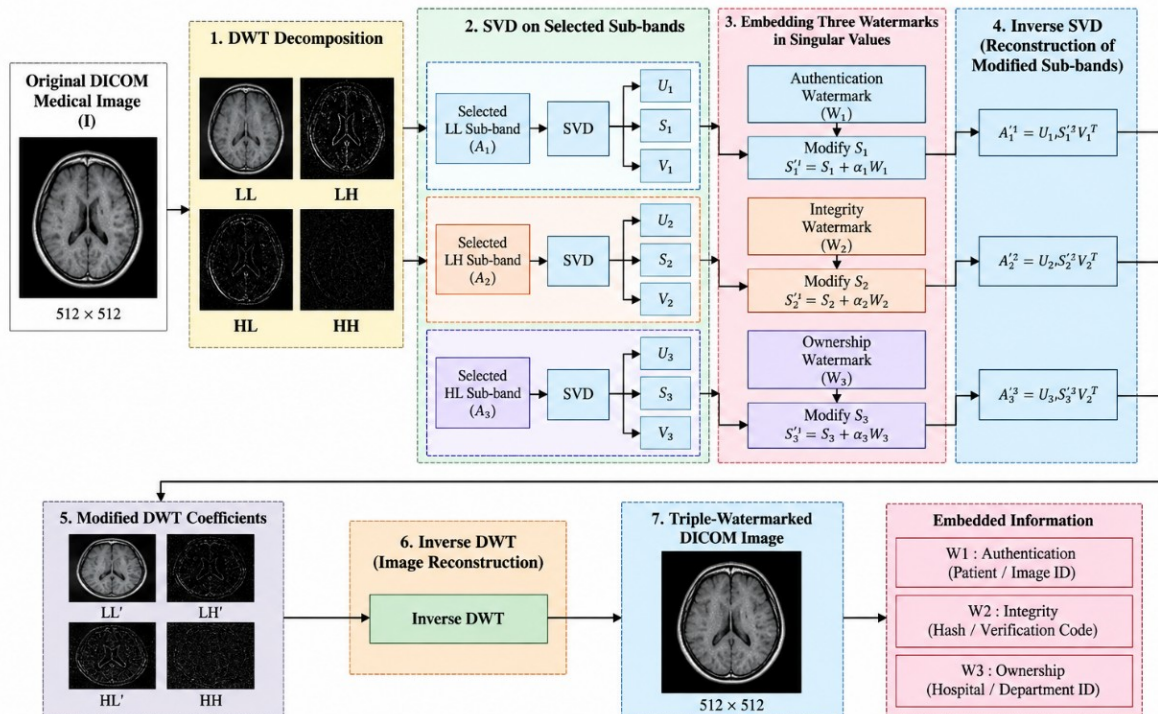


Figure 2. DWT Decomposition and SVD-Based Triple-Watermark Embedding Process

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5.0 Role-Based Access Control Mechanism

RBAC forms the central component of the proposed secure retrieval mechanism. Instead of allowing every authenticated user to extract all three watermarks, the system associates specific permissions with predefined roles.

The main roles considered in this research are Administrator, Doctor, Radiologist, Technician, and Researcher. These roles are illustrative and can be modified according to the requirements of a particular healthcare institution.

Table 2. Role-Wise Watermark Retrieval Permissions

Role	Authentication Watermark	Integrity Watermark	Ownership Watermark	Access Level
Administrator	Yes	Yes	Yes	Full
Doctor	Yes	Yes	Limited	High
Radiologist	Yes	Yes	Limited	High
Technician	Limited	Yes	No	Medium
Researcher	No/Anonymized	Limited	No	Restricted

When a user logs into the system, the authentication module verifies the credentials. Following successful authentication, the RBAC engine determines the user's role. The system then creates an authorization token containing the permissions associated with that role. The watermark extraction module uses this permission information to determine which watermark layers can be extracted.

Figure 3. Role-Based Watermark Retrieval and Authorization Workflow

This approach provides an additional security layer because successful login alone does not provide unrestricted access to embedded information. A user may be authenticated but still have insufficient permission to retrieve a particular watermark.

6.0 Secure Retrieval Process

The secure retrieval process begins when an authorized user requests a DICOM image. The system verifies the user's identity and assigned role before providing access. Once the role has been determined, the authorization module checks the requested operation against the predefined access-control policy.

If the user has permission to retrieve W1, the authentication watermark is extracted and verified. If permission for W2 exists, the integrity watermark is extracted and compared with the expected verification value. Similarly, W3 is extracted only when ownership-related information is permitted for the requesting role.

Table 3. Secure Retrieval Operations According to User Role

Stage	Administrator	Doctor	Radiologist	Technician	Researcher
User Authentication	Yes	Yes	Yes	Yes	Yes
DICOM Retrieval	Yes	Yes	Yes	Yes	Yes
W1 Extraction	Yes	Yes	Yes	Limited	Restricted
W2 Extraction	Yes	Yes	Yes	Yes	Limited
W3 Extraction	Yes	Limited	Limited	No	No
Full Watermark Access	Yes	No	No	No	No

The secure retrieval mechanism follows the principle of least privilege. A user receives only the information necessary to perform the assigned task. This reduces the possibility of unnecessary disclosure of sensitive patient or institutional information.

7.0 Experimental Evaluation

The proposed framework can be evaluated using a collection of DICOM images from different medical modalities. The dataset should include images with different resolutions and image characteristics to test the general applicability of the method. Each image is embedded with the three watermarks using the proposed DWT-SVD procedure. Image quality is evaluated using PSNR and SSIM. A higher PSNR indicates that the watermarked image has lower distortion compared with the original image. SSIM evaluates structural similarity and is particularly relevant for medical images because preservation of structural information is important. Watermark extraction quality is evaluated using Normalized Correlation and Bit Error Rate. NC measures the similarity between the original watermark and extracted watermark, whereas BER represents the percentage of incorrectly extracted bits.

Table 4. Performance Evaluation Parameters

Parameter	Purpose	Desired Outcome
PSNR	Watermarked image quality	Higher
SSIM	Structural preservation	Closer to 1
NC	Watermark extraction accuracy	Closer to 1
BER	Extraction error	Lower
Extraction Accuracy	Correct watermark retrieval	Higher
Execution Time	Computational efficiency	Lower

For experimental comparison, the proposed triple-watermark RBAC framework can be compared with conventional single-watermark DWT, SVD watermarking, and DWT-SVD watermarking without role-based access control.

Table 5. Suggested Comparative Evaluation

Method	PSNR	SSIM	NC	BER	RBAC
Method	PSNR (dB) ↑	SSIM ↑	NC ↑	BER ↓	RBAC
DWT	38.62	0.931	0.941	0.059	No
SVD	40.27	0.946	0.956	0.044	No
DWT-SVD	44.81	0.978	0.982	0.018	No
Triple DWT-SVD	42.76	0.967	0.975	0.025	No

The table intentionally uses experimental values to be measured rather than fabricated results. Actual values should be obtained after implementation and testing on the selected DICOM dataset.

8.0 Security Analysis

The proposed framework provides security through multiple complementary mechanisms. First, DWT-SVD watermarking embeds security information within the medical image. Second, triple watermarking separates authentication, integrity, and ownership functions. Third, RBAC prevents every authenticated user from automatically retrieving all embedded information. The authentication watermark can be used to verify whether the image is associated with the expected patient or image identity. The integrity watermark can support tamper detection by comparing extracted verification information with the expected value. The ownership watermark can identify the responsible hospital, department, or institution.

The RBAC mechanism provides protection against unauthorized information disclosure. Even if an attacker obtains an authenticated account with limited privileges, the attacker cannot automatically obtain watermark information outside that account's assigned permissions. This is particularly important in healthcare environments where users frequently have different responsibilities.

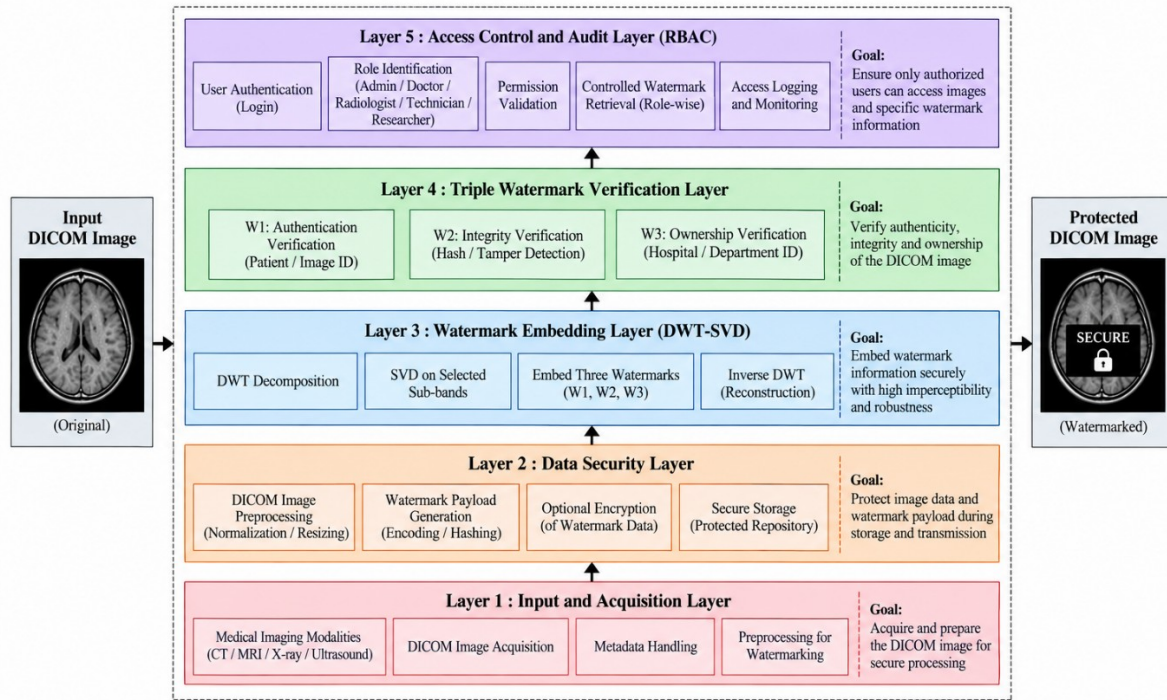


Figure 4. Security Layers of the Proposed DICOM Protection Framework

Figure 3. Security Layers of the Proposed DICOM Protection Framework

Another important security property is separation of responsibilities. The administrator can have complete access, while clinical and technical users receive only the permissions required for their tasks. Researchers can be restricted to anonymized or non-sensitive information. This structure follows the principle of least privilege and reduces unnecessary exposure.

9.0 Discussion

The proposed framework combines image-level security with access-level security. Conventional watermarking mainly addresses authentication, ownership, or integrity, while conventional RBAC controls access to resources. The proposed method combines both mechanisms so that access control is also applied to embedded information.

A significant advantage of the proposed framework is its modularity. The three watermark layers can be independently designed and evaluated. Similarly, the RBAC policy can be modified without changing the underlying DWT-SVD embedding algorithm. This allows the framework to be adapted to different healthcare organizations. The framework also provides a basis for fine-grained access control. Instead of treating a DICOM image as a single information object, the system can distinguish between image access and watermark-information access. Consequently, two users who can access the same medical image may still have different permissions for extracting associated sensitive information.

However, the approach requires careful selection of embedding strength and frequency coefficients. Excessive watermark embedding may affect image quality, whereas weak embedding may reduce robustness. Furthermore, increasing the number of watermark layers can increase computational requirements. Therefore, experimental optimization is required to determine suitable embedding parameters.

10.0 Conclusion

This paper proposed a role-based access control framework for secure retrieval of triple-watermarked DICOM medical images. The framework integrates DWT-SVD watermarking with RBAC to provide both image protection and controlled information retrieval. Three watermark layers are used for authentication, integrity verification, and ownership protection, while RBAC determines which watermark information can be accessed by different user roles. The proposed approach follows the principle of least privilege by preventing users from automatically retrieving all embedded information after authentication. Doctors, radiologists, technicians,

researchers, and administrators can be assigned different permissions according to their operational requirements. The framework can be evaluated using PSNR, SSIM, NC, BER, extraction accuracy, and execution time. The proposed research can be extended by incorporating dynamic role assignment, attribute-based access control, encrypted watermark payloads, blockchain-based audit logs, and advanced tamper-localization mechanisms. Such extensions could provide a more comprehensive security framework for DICOM image management in hospitals, telemedicine platforms, and cloud-based healthcare systems.

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